Can we live on planet 0307M?

You hear the landing mechanism of your spacecraft engage. The loud thrusters incite a ringing in your ears as the shuttle that contains you, your research team, and multiple billion-dollar worth of equipment makes planetfall. The shuttle's double-walled steel and lead doors unlatch as the bay opens. You exit the shuttle and step into one cramped spacesuit boot in front of the other as you ecstatically waddle onto the surface of planet 0307M, where you have been contracted as a part of their habitability team. You are attempting to collect a wide assortment of data to grade planet 0307M's surface environment on a scale from A-F, noting that a low surface habitability grade isn't necessarily detrimental to the planet's overall habitability. A-grade denotes optimal habitability with balanced atmospheric conditions, suitable surface temperatures, and accessible resources. A B-grade suggests promising habitability potential with only a few environmental adjustments needed. Furthermore, a C-grade indicates moderate habitability with significant challenges, potentially including atmospheric issues or scarce resources, while a D-grade signifies limited habitability, requiring substantial modification; a stellarly locked exoplanet may fall into this category. E-grade points to poor habitability with severe environmental constraints, while an F-grade represents conditions that are hostile to human life without dramatic alteration(terraforming). Your developed equipment includes a fleet of rovers equipped with weather stations, a crew of drilling rovers, a large group of sonar drones, a constellation of satellites, and assorted pairs of geophone and seismic rovers.

The weather station rovers are intended to collect a slurry of weather data autonomously, eventually putting together a climate report for the planet, but initially providing a weather report. To autonomously collect weather data over long periods to compile a climate report, the rovers would be powered by Radioisotope Thermoelectric Generators (RTGs) to provide long-term power, with the ideal fuel source, Plutonium-238, having a high heat output and a long half-life. The weather station rovers would have a multitude of sensors and instruments, but the most notable would be a hygrometer to detect moisture in the air. Air moisture could indicate nearby groundwater, so that the moisture reading will be collected along with metadata denoting time and location. A coordinate system would be created for 0307M, aligned with major geological features, to track data collected by rovers and to align satellite geolocation. In locations where significant hygrometer readings were detected, whether on the surface or within surface-level caves, drilling rovers would be sent to collect more data. Aside from the hygrometer, another integral instrument aboard the weather station rovers would be a gas sampling and analysis tool. The tool would include a gas chromatograph, a Quadrupole Mass Spectrometer, and a Tunable Laser Spectrometer (TLS) to harvest gas samples and analyze their composition immediately. It is necessary to take a variety of gas samples from around the planet, since atmospheric composition can change, especially with denser gas impurities lingering near the surface. If a planet is 'stellarly locked', which mainly occurs with exoplanets that orbit red dwarf stars, half of the planet is always receiving light, and the other half never sees it. The side receiving constant stellar radiation could have a completely different atmospheric composition than the side that does not. Gas samples could be analyzed to create an atmospheric report of the planet. Likely, the planet would not be stellarly locked, so it would have a uniform atmosphere. However, if stellarly locked, a Goldilocks zone that receives just the right amount of light on the meridian may still exist. This Goldilocks zone could still be habitable, though it would be an eternal day, pushing it down the grading scale due to the greater effort required for colonization. If carbon dioxide, methane, and water vapor were present in the gas samples, the planet is likely to have a potent greenhouse gas effect, which is essential to its habitability. If these gases are present, 0307M can trap heat from solar radiation inside its atmosphere, allowing the planet to regulate surface temperatures ideal for growth and human life. If the greenhouse gas effect is present, but not at the perfect balance, habitability is still possible. Hydrothermal vents can be harnessed to collect CO_2 and nitrogen to remediate the atmosphere's gas balance.

The drilling rovers are meant to bore a series of holes, scan the stratigraphic column with a camera probe, and then fill in the majority of the excavated borings. Occasional borings will be kept for physical analysis and the study of stratigraphic layers, but autonomous camera probing will allow researchers to create a stratigraphic map of the cross-sections of various areas. Cored soil samples will be analyzed in a mass spectrometer to identify the presence of carbon, hydrogen, oxygen, and nitrogen, elements essential to life, to assess habitability further. For instance, the measurement of these elements directly informs whether crops could be grown or whether local soils would need substantial modification to support agriculture. Understanding the local stratigraphy will provide information about building stability, possible carbonate content, and the relative abundance of fossils in productive layers (assuming there is any). For example, if a site shows uniform, stable layers, it could support the foundations of habitats and infrastructure. By contrast, uneven or brittle substrata can increase construction costs or risk structural failures. Assuming the stratigraphy is not uniform, like the Eagle Ford shale, it would be harder to understand the narrative of planet formation. If the stratigraphy from borings proves relatively uniform, it can help piece together the narrative of the planet's eras, potentially revealing patterns in layers of high critical mineral content. A high critical mineral content could be significant towards raising the habitability score of 0307M. For example, mined copper could be used in reactor assemblies, while mined lithium might be used in battery production for storing the energy produced. Locally mined minerals would decrease 0307M's reliance on earth imports, thereby further enhancing sustainability by reducing dependence on earth's resources. This would make everyday necessities, like energy storage for homes and vehicles, cheaper and more reliable for future inhabitants. Knowing where to begin mining is essential to a successful, efficient operation. Another potential use of the drilling rovers would be borings looking for unstable earth and a lack of it altogether. To elaborate, boring columns can help identify clay layers that are unsafe to build on, as well as the presence of caves and karst features. Locating unstable ground or large underground voids helps prevent dangers such as building collapse or the sudden formation of sinkholes where communities might develop. Assuming there is an active groundwater system, erosion will have taken place, and karstic environments will be present. In terms of habitability, extensive sinkholes pose a significant risk as geological hazards. Luckily, with a wide map of borings and ground samples, sinkholes and underground cave systems could be mapped out, helping future settlers choose safe locations for residences and infrastructure. This stratigraphic data will be integrated with atmospheric readings and topographic findings to provide a comprehensive analysis of the planet's habitability. By synthesizing these datasets, it will be possible to assess not only the surface and subsurface environments (which extend only 100m), but also how they interact, informing the overall habitability assessment more completely.

Conversely, the sonar drones would be sent to collect topographic data for a detailed map of the planet's surface. Topographical analysis would evidence any erosional carving from water, fossilized horseshoe lakes, or faults where groundwater might be present. A scanned fault would be helpful because it would provide evidence of plate tectonics, but it could also be a damage zone, where it is very permeable, and groundwater is likely to be present. The sonar drones would have artificial intelligence software onboard to simultaneously analyze topography and search for concave surfaces to identify anticlines easily. Anticlines serve as a structural trap for oil and gas accumulation, since they will accumulate at the crest under an impermeable layer of rock. One example of how this data could be applied is through the detection of an anticline similar to those found on Earth, which typically signal locations where natural gas or petroleum has accumulated. If the sonar drones map out such an anticline, the drilling rovers can be directed precisely to that site, increasing the likelihood of a successful extraction and minimizing unnecessary drilling elsewhere. Possible gas discoveries could be geologic hydrogen, which would be essential for spacecraft fuel production. In addition, the discovery of oil on 0307M would be a fossil discovery in itself, since oil is composed of the fossilized remains of algae, bacteria, and plankton. Even though sustainability is the goal, a fossil fuel discovery from boring could still yield significant insights into the planet's history and the paleoclimates it once supported, possibly shedding light on future climate patterns as well. Sonar drones could also be sent into discovered cave systems to map their interiors, allowing scientists to create a 3D map of the planet's surface cave network. This process would require some modification to the drones, but it wouldn't be impossible. A mapped-out cave network allows for pre-excavated mining operations that may have already exposed ore nodules, as well as the potential for a geological hazard map. The 3D cave map, superimposed with the stratigraphic map, could create an interactive map interface of the planet where sinkhole hazards could be marked by off-site geologists or artificial intelligence.

Additionally, there would be planet-specific satellites for big-picture surface imaging, as well as for pinging the rovers to collect their data and send it to the space station. For surface habitability purposes, though, the satellites would be used for surface imaging and water and ice detection. With sufficient tectonic evidence, it could help create a map of the planet to study how its surface has changed over time. A narrative of plate tectonics could lead to the discovery of metal deposits present on plate boundaries. For example, subduction zones at convergent plate boundaries, such as those in the Andes Mountains on Earth, concentrate copper through volcanic activity and hydrothermal processes. Superheated hydrothermal fluids heated by magma in subduction zones circulate through the damage zones of the fractured crust, dissolve metals within the fractured subducting plate and the overlying mantle, and rise to deposit economically significant metals. The satellites could also be used to track surface vegetation, if any exists, and the trends could signify potential water sources and arid biomes. High vegetation density could show the location of a seepage spring and its subsequent water source aquifer.

Circling back to stratigraphic analysis, there would be a commissioning of rover fleets deployed in pairs: one working as a seismic pounding rover, while the other as a geophone receptor, communicating to map the subsurface terrain. High loss ratios at certain [planetary-specific] coordinates may indicate groundwater near the surface. Discovering an aquifer, even a brine pool, is essential for habitability. Desalination, however, requires lots of energy, and that is where nuclear power plants would come into play. Since humans will colonize 0307M, it is essential to lay the groundwork for an economy focused not just on growth but also on sustainability. Since the goal is to colonize another planet, it means Earth has been so severely messed up by industrialization; this next chance must be treated with care. The geophone rovers could also be deployed individually as seismographs to further evidence tectonic data, driving along potential fault lines to collect microseismic activity and determine local tectonic activity, data that can later be compiled into a map to understand the long-term landscape of 0307 M better.

In our solar system, Jupiter's immense gravity serves as a 'vacuum cleaner' for larger space debris threats, pulling large objects towards its surface and taking the impacts for the inner solar system. The assumption that 0307M's solar system will have the same threat-neutralizing system isn't automatic, so the satellite imagery will also be used to detect potential meteorite impact sites for additional mining. Sampling meteorites can enable isotope analysis of zirconium, carbon-14, and thorium, potentially allowing the dating of the rocks. Assuming the rocks contain zircon, it is possible to date the meteorite's formation. Alternatively, it's likely to date the impact using argon-argon dating, determining when the radioisotopes began to decay in shocked material and impact-melt rocks. Having a timeline for when a meteor hit the planet could provide a detailed paleoclimate record that helps understand how habitable the planet is, allowing the calculation of impact likelihoods. Assuming the atmosphere is not as thick and protective as Earth's. There is a lack of a vacuum cleaning gravitational body; a missile protection system could be built to neutralize any asteroid threats, working in conjunction with surface missile sites and defense satellites to break up incoming foreign bodies. Hence, they are small enough to be burned up by the existing atmosphere. A stable, relatively dense atmosphere is essential for habitability, with nearly no workarounds. The atmosphere protects from solar radiation, shields from small-scale space debris, and supports human life and agriculture.